

If a training dataset lacks diverse examples the resulting AI may carry hidden biases or make mistakes when faced with unfamiliar situations. A well-constructed training dataset is vital for developing accurate, robust, and reliable machine learning models. The size, diversity, quality, balance and preprocessing of the training data set significantly impact the model's performance, generalization and fairness.

18. Computer vision is a subfield of artificial intelligence that equips machines with the ability to process, analyze and interpret visual inputs such as images and videos. It uses machine learning to help computers and other systems derive meaningful information from visual data.

It sees and interprets images by breaking them down into a grid of numbers (pixels) and using AI to find patterns, edges and shapes layer by layer.

The process works in 4 steps. ① Capturing the image, pre-processing, Extraction features, interpreting and classifying.

Example: Smartphone face unlock they use infrared / front facing cameras to map facial geometry and verify identity.

Medical Imaging Diagnostics AI algorithms analyze X-rays, MRIs and CT scans to highlight any anomalies or early stage tumors that humans might miss with their naked eye.